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Computer Vision — ITAI 1378

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Reflection Journal:

This lab provided a practical introduction to how digital images are represented and manipulated by computers. While I was already familiar with the idea that computers ultimately process numerical information, seeing images broken down into matrices of pixel values made the concept much more concrete. Examining RGB channels, grayscale conversions, and image properties helped me better understand how visual information is stored and processed before any advanced computer vision techniques are applied. One of the most interesting takeaways was realizing how many image transformations can be performed through relatively simple mathematical operations.

The relationship between mathematics and visual effects became especially clear during the brightness, contrast, filtering, and enhancement exercises. I found it interesting that small numerical changes could dramatically affect the appearance of an image. Among the concepts covered in the lab, convolution-based operations were probably the most challenging to fully understand. It was easy to observe the results of blurring, sharpening, and edge detection, but understanding how different kernel values influence those outcomes required more careful analysis. At the same time, these operations helped me appreciate how traditional image processing techniques form the foundation of many modern computer vision systems.

As part of my experimentation, I explored several variations beyond the standard examples. I compared different contrast levels to observe how image details became more or less visible, explored artistic effects generated through combinations of image processing operations, and compared histogram equalization with CLAHE enhancement. These experiments reinforced the idea that even relatively simple adjustments can significantly change the way information is perceived within an image. I found the comparison between histogram equalization and CLAHE particularly interesting because it demonstrated the difference between global and local contrast enhancement.

One of the most valuable aspects of this lab was understanding the connection between traditional image processing and modern AI tools. Before learning more about computer vision, it was easy to view AI image generation and editing systems as entirely separate technologies. However, the Nano Banana discussion and the AI simulation section demonstrated that many AI-powered tools build upon the same fundamental concepts explored in this lab, including filtering, feature extraction, edge detection, image enhancement, and transformation. The difference is that AI systems learn and combine these operations automatically through large amounts of training data rather than relying solely on manually designed filters.

The techniques explored in this lab also helped me better understand real-world computer vision applications. One area that particularly interests me is medical image analysis. Concepts such as image enhancement, feature extraction, and segmentation play an important role in helping healthcare professionals analyze medical images and identify abnormalities. Understanding the mathematical foundations behind image processing makes more advanced topics such as tumor segmentation, diagnostic support systems, and AI-assisted healthcare applications easier to understand. It also highlights the importance of transparency and human oversight when these technologies are used in high-stakes environments.

Overall, this lab changed the way I think about digital photography and image editing. Rather than viewing filters and enhancements simply as visual effects, I now have a greater appreciation for the mathematical processes occurring behind the scenes. The aspect of image processing I would most like to explore further is the intersection of computer vision, healthcare, and responsible AI. I am particularly interested in understanding how modern computer vision systems identify important visual features and how these systems can support human decision-making while remaining accurate, explainable, and trustworthy.